

Wireless LAN Infrastructure

Wi-Fi 6 (802.11ax)

Wi-Fi 6 access points utilise OFDMA and 1024-QAM to achieve higher spectral efficiency. Indoors, a Wi-Fi 6 access point provides coverage to approximately 500 square metres in open office environments. Density planning assumes 50 concurrent devices per radio.

- Coverage: ~500 sq m per AP (indoor, open office)
- Density: 50 concurrent devices per radio

Wi-Fi 7 (802.11be)

Wi-Fi 7 introduces 320 MHz channel widths and Multi-Link Operation (MLO). Theoretical aggregate throughput reaches 46 Gbps with 4K-QAM and 16 spatial streams.

Backhaul Cabling

Access points that support 2.5 Gbps or greater require Cat 6A cabling to the switch port. PoE++ (802.3bt) is required for high-power radios with multiple spatial streams at full duty cycle.